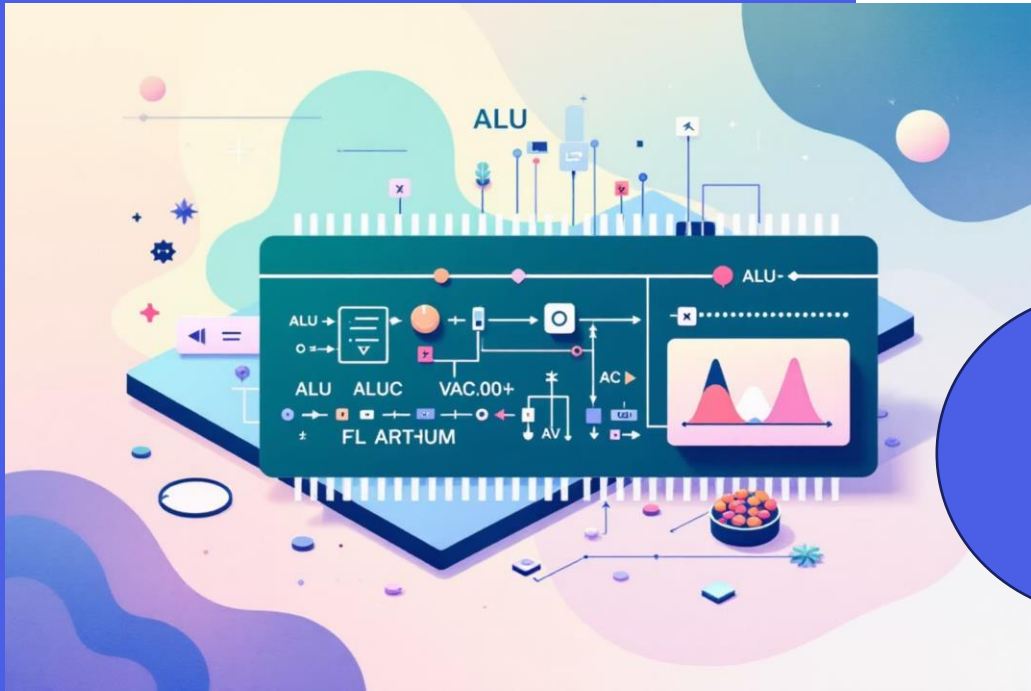


DESIGN AND VERIFICATION OF 4-FUNCTION ALU USING LOGIC GATES

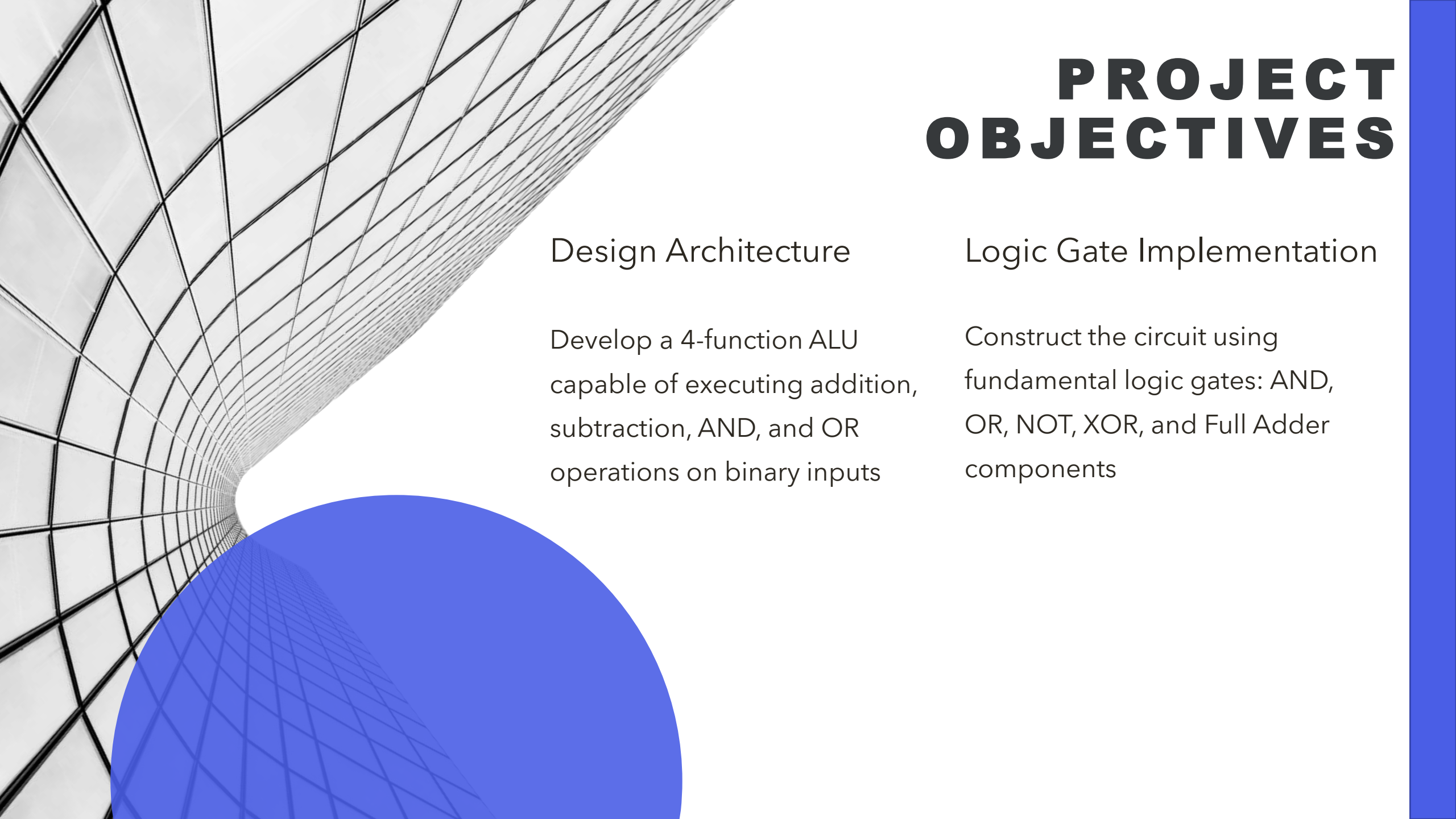
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WHAT IS AN ALU?



The ALU is the computational core of a CPU, executing arithmetic and logical operations on binary data. It forms the foundation of processor architecture and computer system functionality.

Processes instructions by performing calculations, comparisons, and data manipulations essential for program execution and system operations.



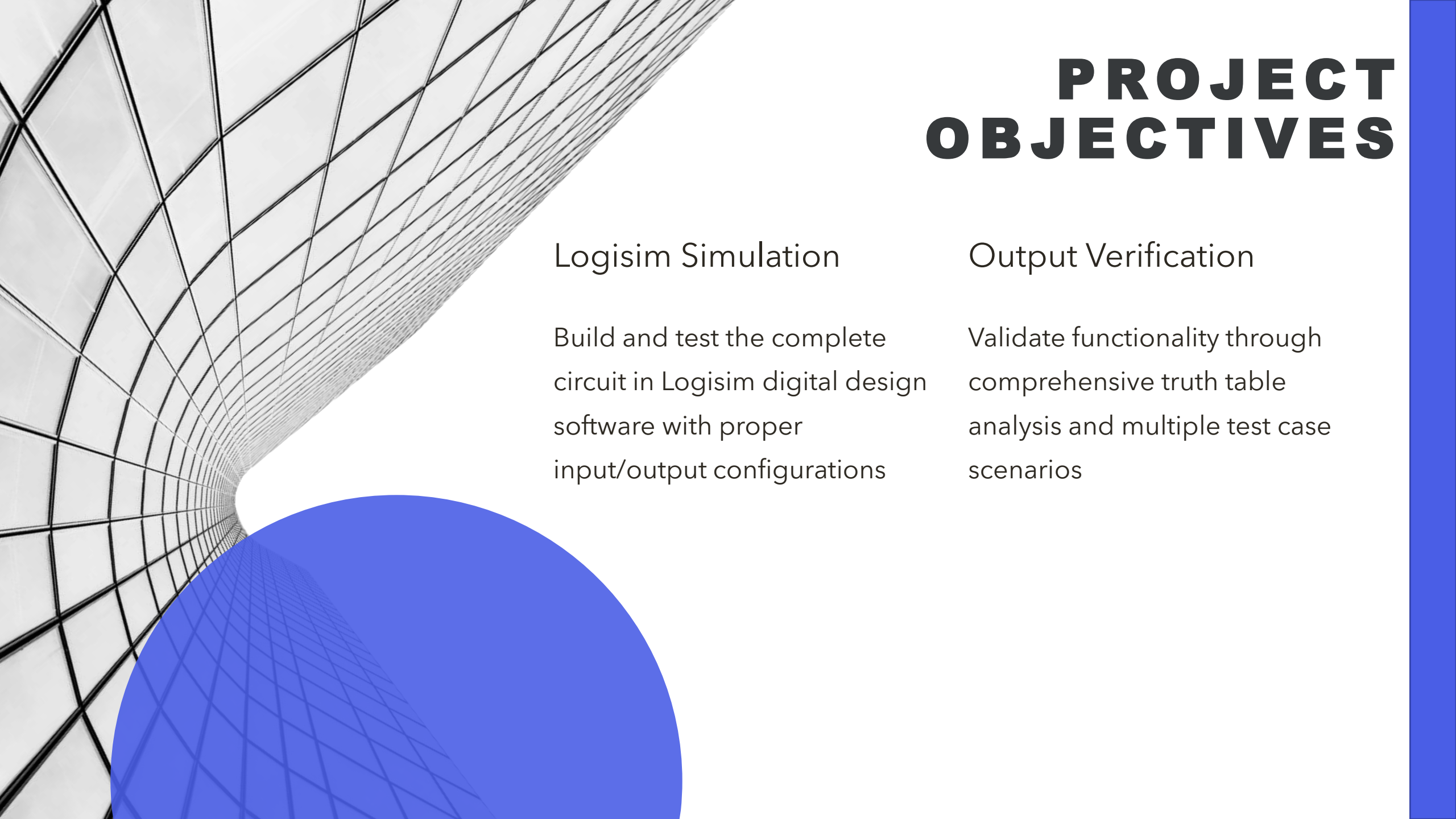
PROJECT OBJECTIVES

Design Architecture

Develop a 4-function ALU capable of executing addition, subtraction, AND, and OR operations on binary inputs

Logic Gate Implementation

Construct the circuit using fundamental logic gates: AND, OR, NOT, XOR, and Full Adder components



PROJECT OBJECTIVES

Logisim Simulation

Build and test the complete circuit in Logisim digital design software with proper input/output configurations

Output Verification

Validate functionality through comprehensive truth table analysis and multiple test case scenarios

COMPONENTS AND BUILDING BLOCKS

LOGIC GATES

AND, OR, NOT, XOR gates for Boolean operations

FULL ADDER

Handles binary addition with carry propagation

MULTIPLEXER

Selects between multiple operation outputs

I/O DEVICES

Input switches and LED displays for testing

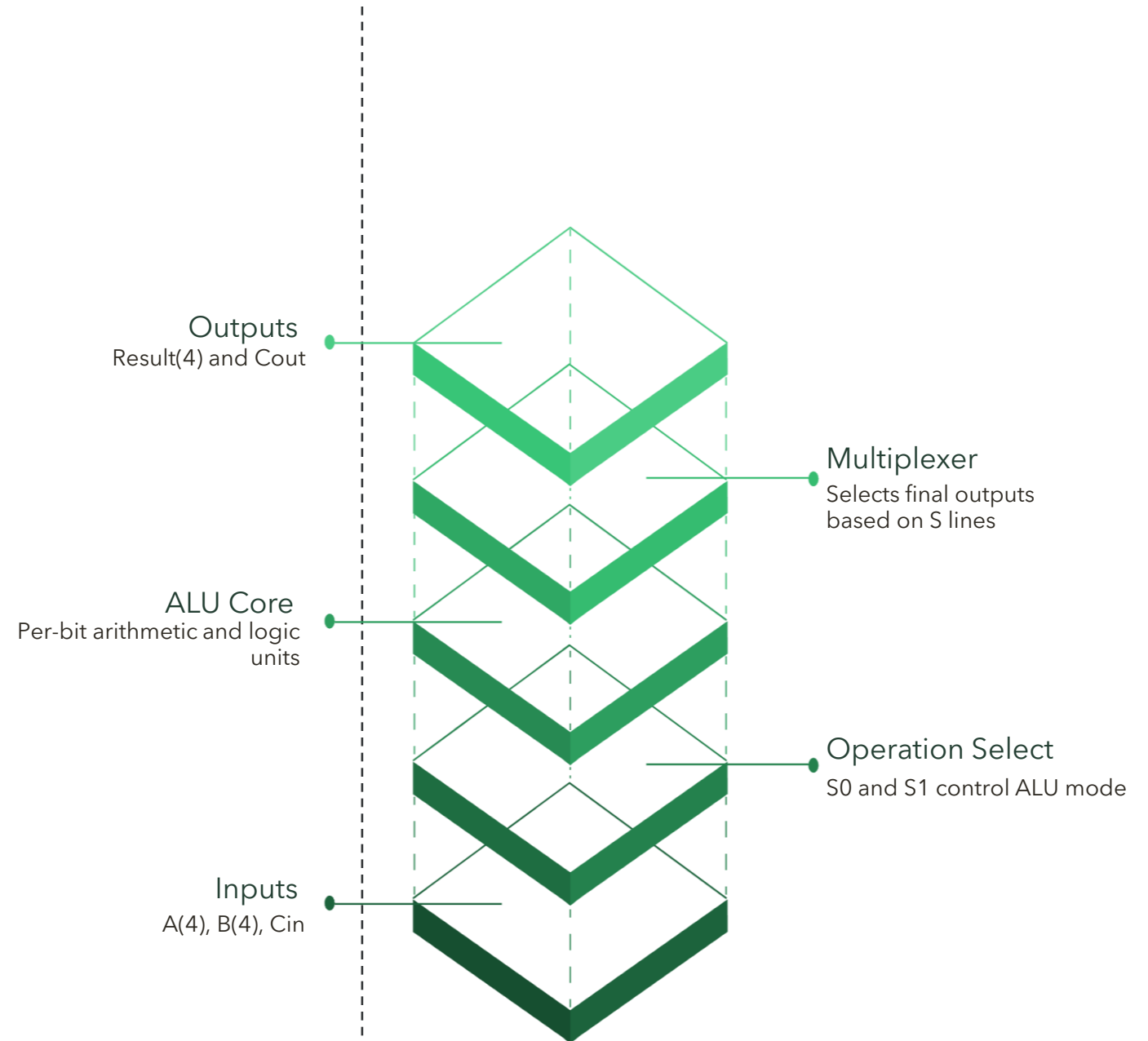
ALU BLOCK DIAGRAM

Input Signals

- Two 4-bit binary inputs (A, B)
- Carry input (Cin)
- Two operation select lines (S0, S1)

Output Signals

- 4-bit result output
- Carry output (Cout)
- Operation selection control



Four Operations Implemented

00

Addition ($A + B$)

Binary addition using Full Adder circuits with carry propagation through all bit positions

01

Subtraction ($A - B$)

Two's complement subtraction implemented using addition with inverted subtrahend

10

AND Operation

Bitwise logical AND between corresponding bits of inputs A and B

11

OR Operation

Bitwise logical OR between corresponding bits of inputs A and B

CIRCUIT DESIGN AND OPERATION

Implementation Details

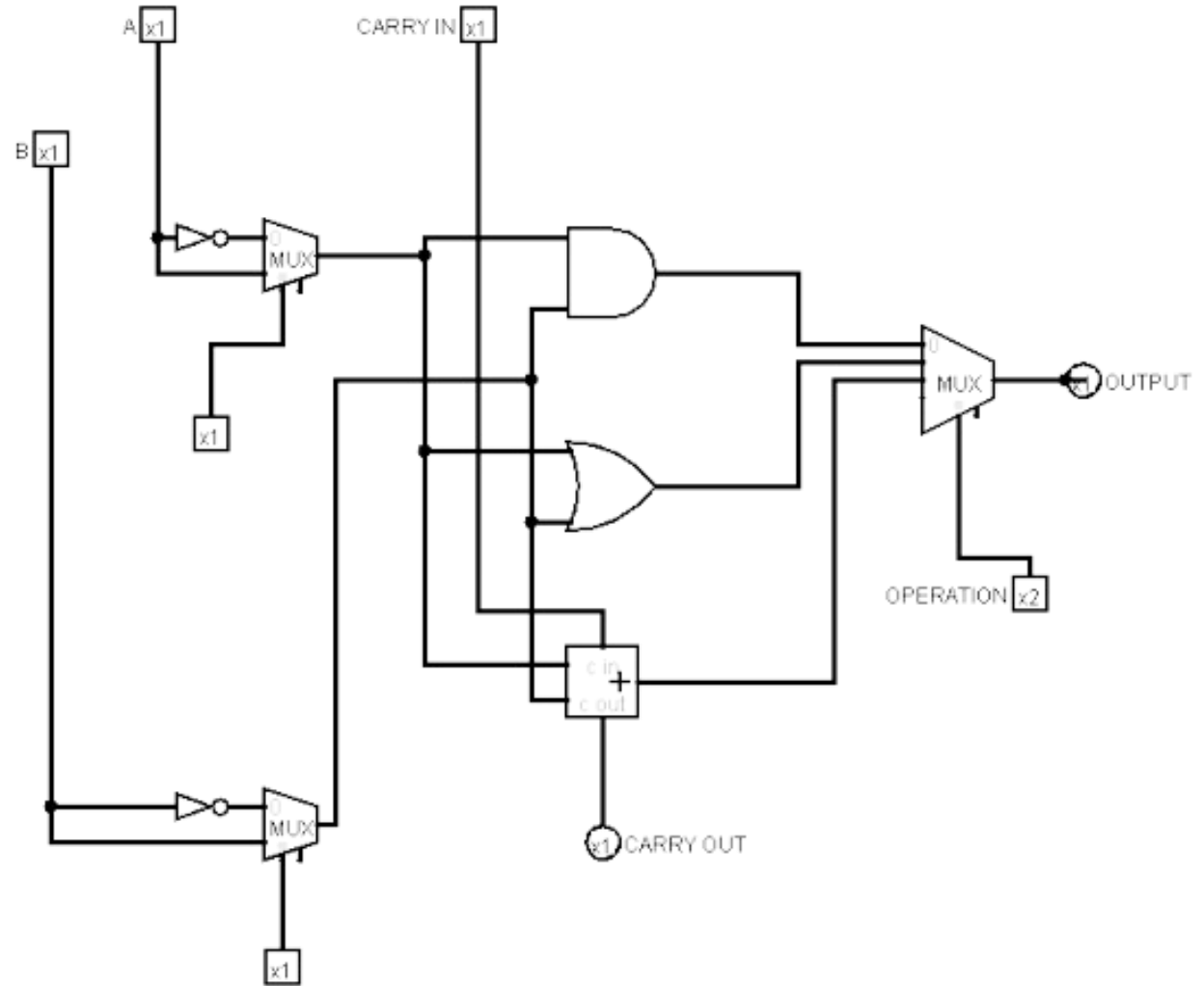
The ALU was constructed in Logisim using discrete logic gates connected through a multiplexer-based selection system.

Control Mechanism

Selection lines S0 and S1 determine the active operation. The multiplexer routes the appropriate operation result to the output based on these control signals.

Carry Logic

Carry propagation implemented through Full Adder circuits, with carry-out from one bit feeding carry-in to the next higher bit position.



TRUTH TABLE VERIFICATION

Operation selection and expected outputs for all four functions:-

S1	S0	OPERATION	OUTPUT
0	0	ADDITION	$A+B$
0	1	SUBTRACTION	$A-B$
1	0	AND	$A \wedge B$
1	1	OR	$A \vee B$

Each operation verified with multiple input combinations to ensure correct functionality across all bit positions.

Testing and Verification Methodology

Individual Gate Testing

Verified functionality of each logic gate and Full Adder circuit independently before integration

Operation Mode Testing

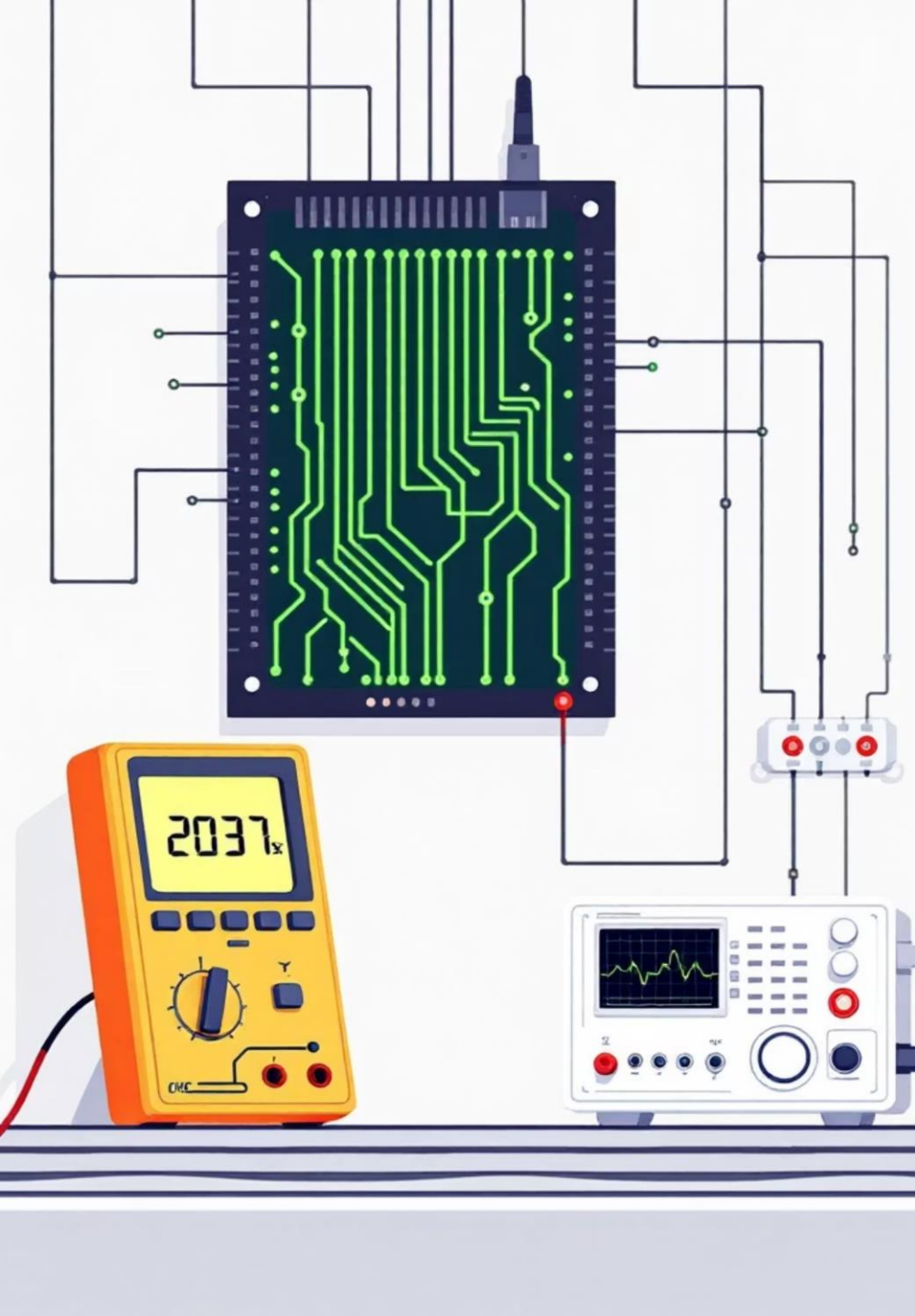
Tested each operation (addition, subtraction, AND, OR) with various input combinations to validate correctness

Edge Case Analysis

Examined overflow conditions, negative results, and boundary values to ensure robust operation

Truth Table Validation

Compared simulated outputs against theoretical truth table predictions for comprehensive verification



Conclusion and Future Scope

Project Success

Successfully designed and implemented a 4-function ALU using fundamental logic gates with complete operational verification through simulation

Key Learning Outcomes

Enhanced understanding of digital logic design principles, circuit construction techniques, and simulation methodologies using industry-standard tools

Future Enhancements

Expand to 8-bit or 16-bit architecture, incorporate additional operations (XOR, shift), implement pipelining, or integrate with CPU control unit for complete processor simulation



THANK YOU

1400. Krishna Kamlesh Singh

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